

SIM2.2 — Channel-Length Modulation in the 035 NMOS

Analog IC Design | LTspice 26.0.2 | 035 LTspice library

1. Simulation setup

The simulation uses the supplied NM Level 8 BSIM3 NMOS model and nmos_035 symbol. Source and bulk are grounded. W/L is held at 10 for five channel lengths: 1, 2, 5, 10, and 20 μm .

Target overdrive is $V_{OV} = 0.500\text{ V}$. Since the model effective threshold varies with geometry, LTspice operating-point V_{th} was read at $V_{DS} = 1.0\text{ V}$, then each gate bias was set to $V_{GS} = V_{th} + V_{OV}$. The calibration netlist is **SIM2_2_NMOS_VTH.cir**. Each gate voltage stays fixed during the output sweep; VOV is matched at the 1.0 V reference point, while the model may shift Vth slightly as VDS changes.

L (μm)	W (μm)	Vth (V)	VGS (V)	VOV (V)
1	10	0.536	1.036	0.500
2	20	0.518	1.018	0.500
5	50	0.506	1.006	0.500
10	100	0.502	1.002	0.500
20	200	0.500	1.000	0.500

VDS is swept from 0.6 V to 2.0 V in 10 mV steps. The sweep begins above the target 0.5 V overdrive, so the plotted range remains in saturation.

2. Extraction method

Using the assignment approximation $I_D \approx I_{D0}(1 + \lambda V_{DS})$, estimate the slope and intercept from the endpoint currents:

$$g_{ds} \approx [I_D(2.0\text{ V}) - I_D(0.6\text{ V})] / 1.4\text{ V}$$

$$I_{D0} \approx I_D(0.6\text{ V}) - 0.6g_{ds}$$

$$\lambda \approx g_{ds} / I_{D0}$$

Output resistance is calculated at $V_{DS} = 1.3\text{ V}$ using the requested formula $r_o \approx 1/[\lambda I_D(1.3\text{ V})]$.

3. Results

L (μm)	W (μm)	ID @ 1.3 V (μA)	gds (μS)	λ (V^{-1})	ro (k Ω)
1	10	158.037	3.489	0.022876	276.611
2	20	165.261	2.464	0.015263	396.453
5	50	169.485	1.729	0.010358	569.653
10	100	170.948	1.379	0.008162	716.706
20	200	171.633	1.097	0.006455	902.644

4. Output characteristics

Each trace is the drain current for a separate L/W pair. The upward slope in saturation shows channel-length modulation; longer-channel curves are flatter.

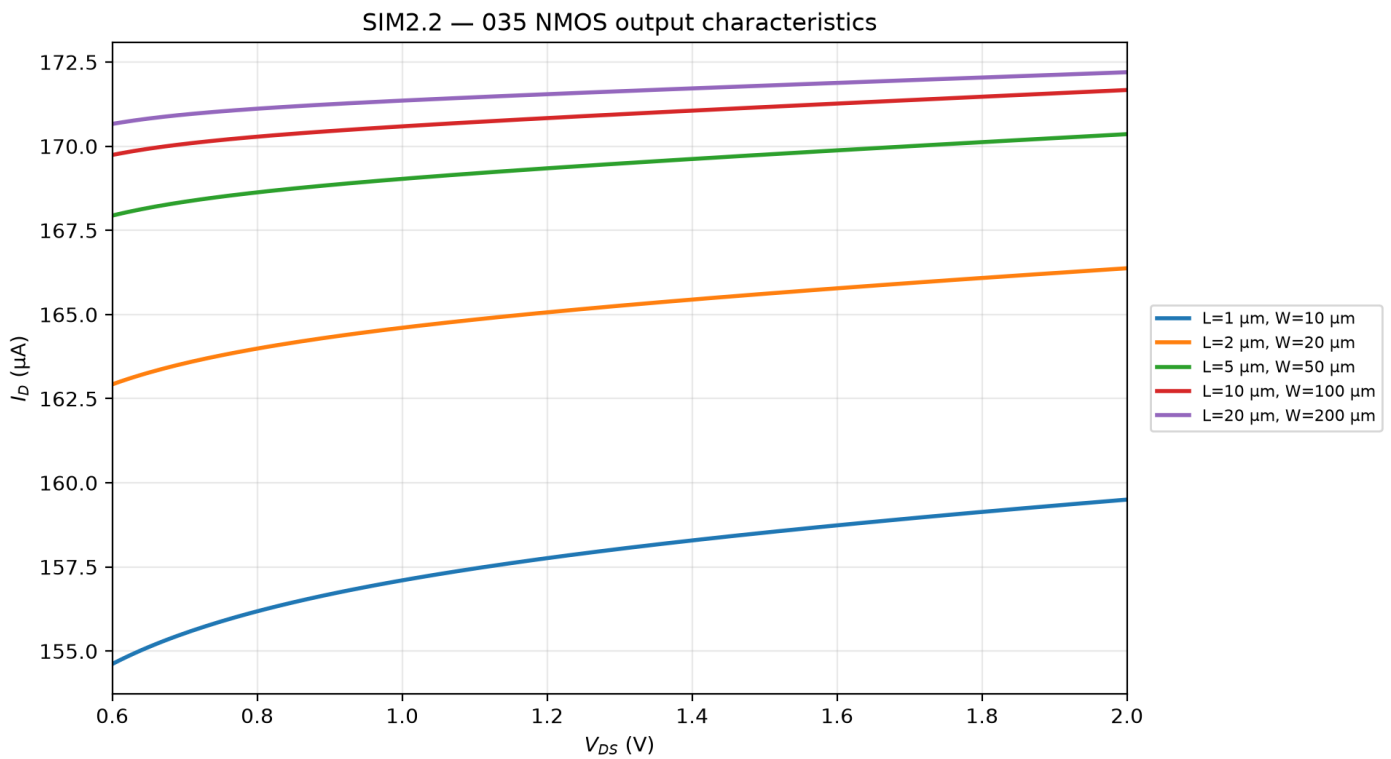


Figure 1. I_D versus V_{DS} for five 035 NMOS geometries. Every device uses $W/L = 10$ and calibrated $V_{OV} \approx 0.5$ V.

The five curves are nearly linear over the selected saturation region. Their slopes decrease as L increases, producing a smaller extracted λ .

5. λ and r_o versus channel length

SIM2.2 — 035 NMOS, $W/L = 10$, $V_{OV} = 0.5$ V

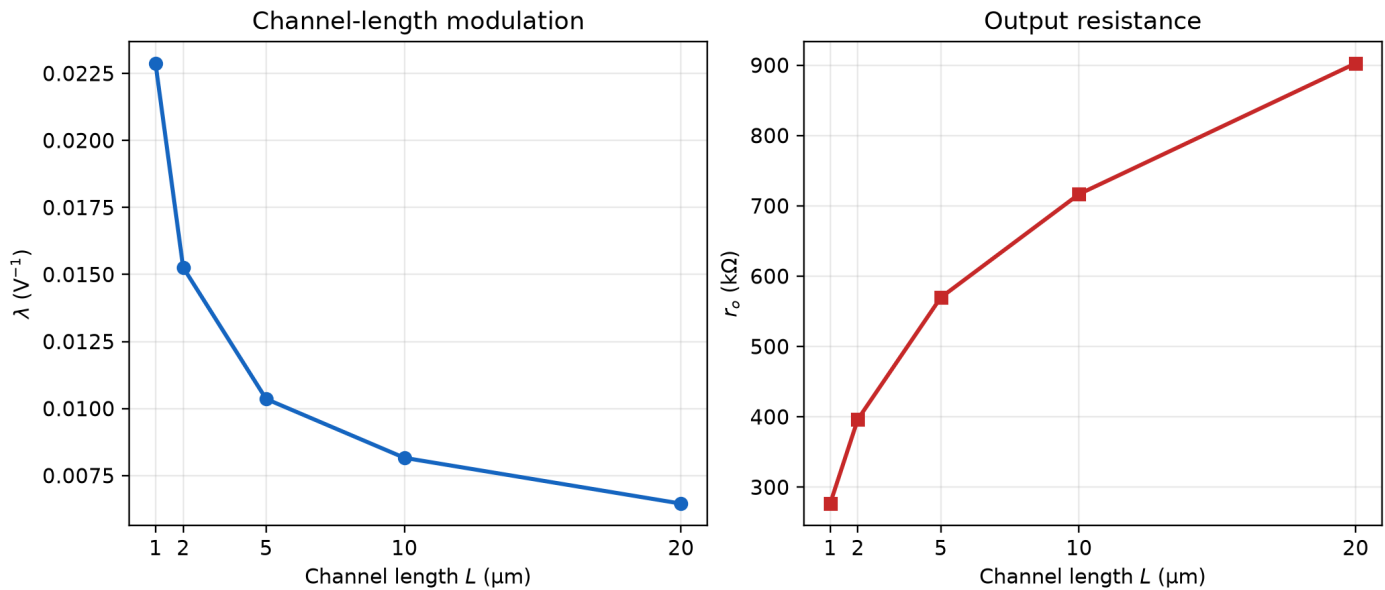


Figure 2. Extracted channel-length modulation parameter and output resistance.

As L increases from 1 μm to 20 μm , λ decreases from $0.022876 V^{-1}$ to $0.006455 V^{-1}$, while r_o increases from 276.6 $k\Omega$ to 902.6 $k\Omega$. A longer channel reduces the relative effect of V_{DS} on channel length, producing a flatter output curve and higher output resistance.

ID at 1.3 V rises modestly from 158.0 μA to 171.6 μA despite constant W/L and calibrated overdrive. This reflects the supplied BSIM3 model's geometry-dependent behavior; the ideal long-channel square-law model would predict less variation.

Reproducible LTspice sources: SIM2_2_NMOS.asc (output sweep) and SIM2_2_NMOS_VTH.cir (operating-point calibration). CSV data files are included beside the report.